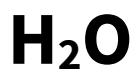


2.A Atoms and the Periodic Table I

Chemical Symbols. Matter is made of atoms—this simple idea pervades modern science, but implications of the idea are profound. Here, we introduce some fundamental conventions for representing atoms and collections of them.

- **Atoms** are the fundamental building blocks of matter—particles that give matter its identity. While we now know that atoms can be divided into subatomic particles, such divisions are beyond the scope of the atom. To the chemist, atoms remain indivisible. (However, we will frequently “look inside” atoms.)
- **Scanning tunneling microscopy (STM)** and **atomic force microscopy (AFM)** can be used to image at the atomic level.
- The chemical elements consist of a single type of atom and are represented using a **chemical symbol** of one or two letters. For example, the symbol “Mg” may represent:
 - A single atom of magnesium
 - A macroscopic sample of magnesium metal
- We combine chemical symbols with subscripted numbers, parentheses, and other symbols to create **chemical formulas**.
 - Subscript numbers in formulas indicate the number of atoms of each element present.
 - Parentheses are used to group atoms linked by covalent bonds (e.g., polyatomic ions).
- What atoms are present in the substances represented below?



Laws of Chemical Combination. Nowadays, we take the atomic theory almost as a given. However, it wasn't always so. Observations of the nature and behavior of elements and compounds provide strong support for the atomic theory.

- **The Law of Conservation of Mass:** the mass of material present at the end of a chemical reaction is equal to the mass of material present at the start of the reaction.
 - During chemical change, matter is neither created nor destroyed.
 - The identities of atoms do not change during a chemical reaction.
- **The Law of Definite Proportions:** every compound has a constant set of proportions by mass of the elements in that compound, which define it.
 - Two different compounds *may* contain the same mass proportions of the elements (for example, NO_2 and N_2O_4).
 - However, two substances with different mass proportions of the elements *must* be different compounds.
- **The Law of Multiple Proportions:** different compounds may contain the same elements in different mass proportions. In that case, those mass proportions are related to each other by small, whole-number ratios.
 - In chemical formulas, subscripts must be positive integers. There is no such thing as “half an atom.”
 - Mass proportions of the same two elements can only differ by the mass of a whole atom.

Dalton's Atomic Theory. Through a good portion of the course (prior to the introduction of the quantum model of the atom in chapter 8), Dalton's theory will form our conception of the nature of atoms. Dalton's theory has four tenets:

1. Matter is made up of very tiny, indivisible particles called atoms.
2. Each atom of a particular element has the same mass, but the mass of an atom of one element is different from the mass of an atom of any other element.¹
3. Atoms combine to form what we now call molecules. When they do so, atoms combine in small, whole-number ratios.
4. Atoms of some pairs of elements can combine with each other in different, small, whole-number ratios to form different compounds.

¹ Dalton was unaware of the existence of **isotopes**, atoms of the same element with different mass. We'll tweak this postulate after introducing isotopes shortly.

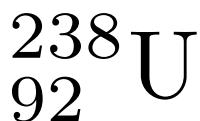
The Classical (Planetary) Model of the Atom. Experiments in the nineteenth and early twentieth centuries established the existence of particles within the atom and the organization of these particles into a highly concentrated central nucleus of positive charge and a surrounding region containing negative charge.

- The **electron**, a negatively charged constituent of the atom (**subatomic particle**), was discovered by J. J. Thomson in the late 1800s.
 - The electron is extremely light, representing only 0.054% of the mass of a hydrogen atom. Electrons can be lost or gained by atoms with relative ease.
 - The “non-electron” part of the atom must have a charge with positive sign equal in magnitude to the charge of the electrons, such that the atom as a whole is electrically neutral.
- The **nucleus** was discovered by Rutherford following his famous gold-foil experiment.
 - Positive charge in the atom resides in the **nucleus**. The nucleus contains highly massive **protons** with positive charge and neutral **neutrons**.
 - Most of the atom is empty space. Light, negatively charged **electrons** are spread throughout this empty space.

Particle	Charge (e)	Mass (u)	Location
Proton	+1	1.0073	In the nucleus
Neutron	0	1.0073	In the nucleus
Electron	-1	5.49×10^{-4}	Outside the nucleus

The elementary charge e is 1.6×10^{-19} Coulombs. One atomic mass unit (u or amu) is 1.66×10^{-24} grams.

- **Atomic number (Z)** is equal to the number of protons in the nucleus of the atom (or, equivalently, the number of electrons in the neutral atom). Atomic number defines the elements—all atoms of a given element have the same atomic number.
- **Mass number (M)** is equal to the sum of protons and neutrons in the nucleus of the atom. Two atoms of the same element may have different mass numbers!
- Superscript and subscript numbers are used in **atomic symbols** to show M and Z , respectively.



Isotopes and the Nature of Bulk Samples of the Elements. If all atoms of an element have the same mass, it follows that atomic masses must all be integer multiples of the mass of a hydrogen atom. *This is not observed.* However, the observed masses of atoms can be readily explained by allowing different atoms of the same element to contain different numbers of neutrons in their nuclei.

- **Isotopes** are atoms of the same element with different numbers of neutrons in the nucleus. Isotopes have the same atomic number (Z) but different mass numbers (M).



- Bulk (macroscopic) samples of the elements and compounds that contain the elements frequently contain a mixture of isotopes. When thinking about bulk samples, we *always* make use of average atomic masses.

2.1. Indicate the numbers of protons, neutrons, and electrons in each of the following isotopes. Keep in mind that the atomic numbers of the elements can be found on the periodic table.

Isotope	Protons	Neutrons	Electrons
Uranium-238			
Lithium-7			
Gold-198			
Oxygen-18			

2.B Atoms and the Periodic Table II

Isotopes Continued. We noted above that bulk collections of atoms consist of a mixture of isotopes and that we use average atomic mass when thinking about bulk samples. Here, we will define a useful unit for expressing atomic masses and precisely define the terms isotopic mass and average atomic mass.

- The **atomic mass unit (u or amu)** is defined as $1/12^{\text{th}}$ the mass of a carbon-12 atom.
 - Atomic masses are on a *relative scale*: carbon-12 is assigned a mass of exactly 12.00 u and all other atomic masses are determined relative to carbon-12.
 - One atomic mass unit is equivalent to 1.66×10^{-24} grams.
- We will use the term **isotopic mass** to refer to the relative mass of an atom of a specific isotope.
- Isotopic masses are not whole numbers due to the **binding energy** of the nucleus. Some portion of the total mass of all nucleons “goes into” the nuclear binding energy.
- The **average atomic mass (average atomic weight)** for an element is a weighted average of all the isotopic masses for that element, with the weights equal to the **abundances** of the isotopes. For an element with N isotopes, each with abundance a_i and isotopic mass m_i ,

$$\bar{m} = \sum_i^N a_i m_i$$

- These are the masses we see on the periodic table.

2.2. There are three naturally occurring isotopes of silicon (atomic masses are listed below). The percent abundance of ^{29}Si is 4.68%. Find the percent abundance of ^{30}Si given the average atomic mass of Si (28.085 u).

$^{28}\text{Si} = 27.977 \text{ u}$

$^{29}\text{Si} = 28.977 \text{ u}$

$^{30}\text{Si} = 29.974 \text{ u}$

Ions. Atoms may contain unequal numbers of protons and electrons and possess a net charge as a result. Charged atoms or molecules are called **ions**.

- **Cations** contain more protons than electrons and have net **positive charge**.
- **Anions** contain more electrons than protons and have net **negative charge**.
- Ions can either be **monatomic**, containing a single atom, or **polyatomic**, consisting of multiple atoms linked by covalent bonds.

2.3. Determine the number of protons, neutrons, and electrons in the ion represented below. Refer to the periodic table as needed!



The Periodic Table. The chemical elements display regular patterns in their properties, bonding, and chemical reactivity. Organization of the elements into the periodic table helps us quickly draw analogies between related elements.

- The **periodic table** is an organization of elements by increasing atomic number with elements having similar chemical properties aligned in vertical columns.
 - Columns of the table are called **groups**;
the groups are numbered 1 – 18 from left to right.²
 - Rows of the table are called **periods**;
the periods are numbered starting with 1 at the top.
- Important classifications of elements on the periodic table:
 - Metals / metalloids / nonmetals
 - Main-group elements / transition metals / inner transition metals
 - Quantum-chemical blocks: *s* / *p* / *d* / *f*
- The inner transition metals (**lanthanides** and **actinides**) are between groups 2 and 3. Which “end” of the series goes under group 3 is debatable, but I like the way ptable.com punts on the issue (see the next page).

² Older numbering systems use Roman numerals and the letters A and B (the latter distinguish between the main-group and transition elements). Modern chemists have pretty much killed the old conventions and you will not see them used in CHEM 1211K. See [this article](#) for more information.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122														7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305														15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798	
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29	
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71																
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103																
				72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)	
				104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)	
				For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.															
				57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97	
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)	

Ptable

Ptable

- Historical group names you should be aware of include the **alkali metals** (group 1), **alkaline earth metals** (group 2), **chalcogens** (group 16), **halogens** (group 17), and **noble gases** (group 18).
-

2.4. Classify each of the following elements as a metal, nonmetal, or metalloid. For the metals, distinguish between main-group metals and transition metals.

Si

Ne

Fe

Sr

Br

Chapter Summary. Some of the amazing things we can now do with atoms and the periodic table...

1. Represent atoms and molecules using chemical formulas.
2. Interpret chemical formulas to count atoms.
3. Apply the laws of chemical combination to define and distinguish between compounds and establish fundamental constraints on the nature of matter.
4. Visualize and count the subatomic particles (protons, neutrons, and electrons) found within neutral atoms and ions.
5. Recognize isotopes and apply isotopic masses and abundances to carry out calculations involving average atomic mass.
6. Describe, classify, and relate elements based on their positions on the periodic table.